Guideline for preparing Organic Teaching Lab Report

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Short Report	Long Report
1. Title 2. Aim 3. Results 4. Discussion 5. Conclusion 6. References 7. Data and Appendix	1. Title 2. Abstract 3. Introduction 4. Procedure 5. Results 6. Discussion 7. Conclusion 8. References 9. Data and Appendix

The report conveys what you have done in a concise, organized, and easy-to-read fashion.

- **Title:** A brief, clear description of the subject of the report.

- **Aim:** A few statements that describe the purpose of the experiment and the method(s) involved. Can include specific reactions, compounds, or properties investigated.

- **Abstract:** A summary the major findings of your experiment. It is usually the last section to be written.

- **Introduction:** A statement of the experiment to be performed and its purpose should be provided. Providing some information on your experiment, such as scientific background, to interest the readers.

- **Procedure:** A clear and concise description of your procedures based on your own experiment records. Note that you should provide the actual mass/volume of reagents and solvents used in your experiment, and not copy from the lab manual. The main goal of this section is to ensure reproducibility of your experiment.

  Sample of procedure: extract from J. Org. Chem. 2014, 79, 11383-11394:

  Synthesis of 2-(3-Methylbut-3-en-1-yl)phenol ***(6a)***
  In a dry round-bottom flask, methyltriphenylphosphonium bromide (4.28 g, 12 mmol) was suspended in freshly distilled toluene (10 mL). The mixture was cooled in an ice bath, and potassium tert-butoxide (1.34 g, 12 mmol) was added slowly. The reaction mixture was stirred at room temperature for 2 h under nitrogen atmosphere. The mixture was cooled to 0 °C, and a solution of 5a (0.458 g, 2.78 mmol) in 5 mL of toluene was added. The reaction was stirred at

room temperature overnight. Upon completion of the reaction (confirmed by TLC), saturated aqueous NH_4Cl solution (10 mL) was added. The mixture was filtered through a pad of Celite with excess EtOAc. The layers were separated, and the aqueous phase was washed with EtOAc (2 × 20 mL). The combined organic phases were washed with brine, dried over sodium sulfate, filtered, and concentrated in vacuo. The crude product was purified by flash column chromatography to afford phenol 6a as a yellow oil (0.450 g, 0.278 mmol, quantitative yield), Rf = 0.5 (1:9 EtOAc/Hex).

- **Results:**
  a. Yield calculation.
  b. Analyse the data: typically for an organic experiment, you will need to analyse the ^{1}H NMR spectrum. A good way is to present this data in a table with clear assignments of the signals. Identify any impurities if possible.
  c. Other possible data includes ^{13}C NMR, IR, MS, melting point, boiling point, etc. when applicable.

  Examples of data reporting in a research paper: for compound 6a – for a teaching report, you should use table format with clear assignment.

  ^{1}H NMR (500 MHz, CDCl3) δ 7.24 (dd, J = 7.3, 1.5 Hz, 1H), 7.17 (td, J = 7.8, 1.5 Hz, 1H), 6.98 (td, J = 7.3, 1.0 Hz, 1H), 6.83 (dd, J = 7.8, 1.0 Hz, 1H), 5.38 (br s, 1H), 4.88–4.86 (m, 2H), 2.89–2.86 (m, 2H), 2.44–2.41 (m, 2H), 1.89 (s, 3H).
  ^{13}C NMR (125 MHz, CDCl3) δ 153.2, 145.8, 130.0, 128.2, 127.1, 120.8, 115.2, 110.1, 37.6, 28.4, 22.6. IR (thin film) 3424, 3072, 1648, 1236. HRMS (ESI) calcd for $[C_{11}H_{14}O - H]^-$ 161.0972, found 161.0968.

- **Discussion:** This is the most important section of the report. It demonstrates how you understand and interpret the data collected. You should:
  a. Briefly summarize the key results of each experiment and answer any questions that are stated in the lab manual when applicable.
  b. Explain any difficulties or problems which may affect the results.
    i. Mechanism: What by-products can be formed, and how do the reaction conditions affect the distribution of the product and by-products?
    ii. Method errors: For example, incompleteness of reactions due to time constraints, the instability of species, and possible interference.
    iii. Gross errors: occur infrequently and often result from an experimental blunder, such as misreading a scale or interpreting a number incorrectly.

  c. Suggest how the experimental procedure or design could be improved with clear literature reference(s). For example, in the literature, the reaction may be

carried out in different reaction condition(s) like solvent, temperature, time, and/or reagent like different reducing agents such as $NaBH_4$, $LiBH_4$, $LiAlH_4$.

Tip for good discussion: Write a preliminary outline of the points to be discussed. This approach keeps the report flow logical and prevents rambling.

- **Conclusion:** Summarise your experiment results in a few sentences. Did you achieve the objective(s) of the experiment? Highlight important results, i.e., was the desired product obtained in good yield and/or purity?

- **References:** A list of references to journal articles, books, patents, etc. (Note: make sure that you use only primary sources). You can refer to the ACS (American Chemistry Society) style guide on NUS library website: https://libguides.nus.edu.sg/c.php?g=145626&p=955284

- **Data and Appendix:** Any raw data (^{1}H-NMR, IR spectra) or datasheet should be attached behind the document.

Tip: write the report in the following sequence: Procedure → Results → Discussion → Introduction → Conclusion/ Abstract/ Aim.

For more tips and examples on how to write an Organic teaching lab report, you can refer to the online resources in the reference section.

References:

Coghill, A. M.; Garson, L. R. *The ACS Style Guide: Effective Communication of Scientific Information*; 2006.

Spector Thomas, “Writing a Scientific Manuscript. Highlights for Success,” J. Chem. Educ. **1994**, 71, 47-50.